L3 Direction Fields

Section 1.3

In many real-world applications, it will be impossible to find an explicit solution to the DE considered. So we must rely on other methods to analyze or approximate the solution. A useful strategy for 1st order ODEs is to sketch the direction field for the ODE.

When an ODE is written in the form

$$\frac{dy}{dx} = f(x,y), \quad \text{order } 1$$

it provides the slope of the solution at any (x,y).

<u>**Def.**</u> A **direction field** is a plot of short line segments drawn at various points in the xy-plane showing the slope of the solution curve at that point.

The direction field suggests the appearance of a family of solution curves of the DE making it possible to visualize aspects of the solution.

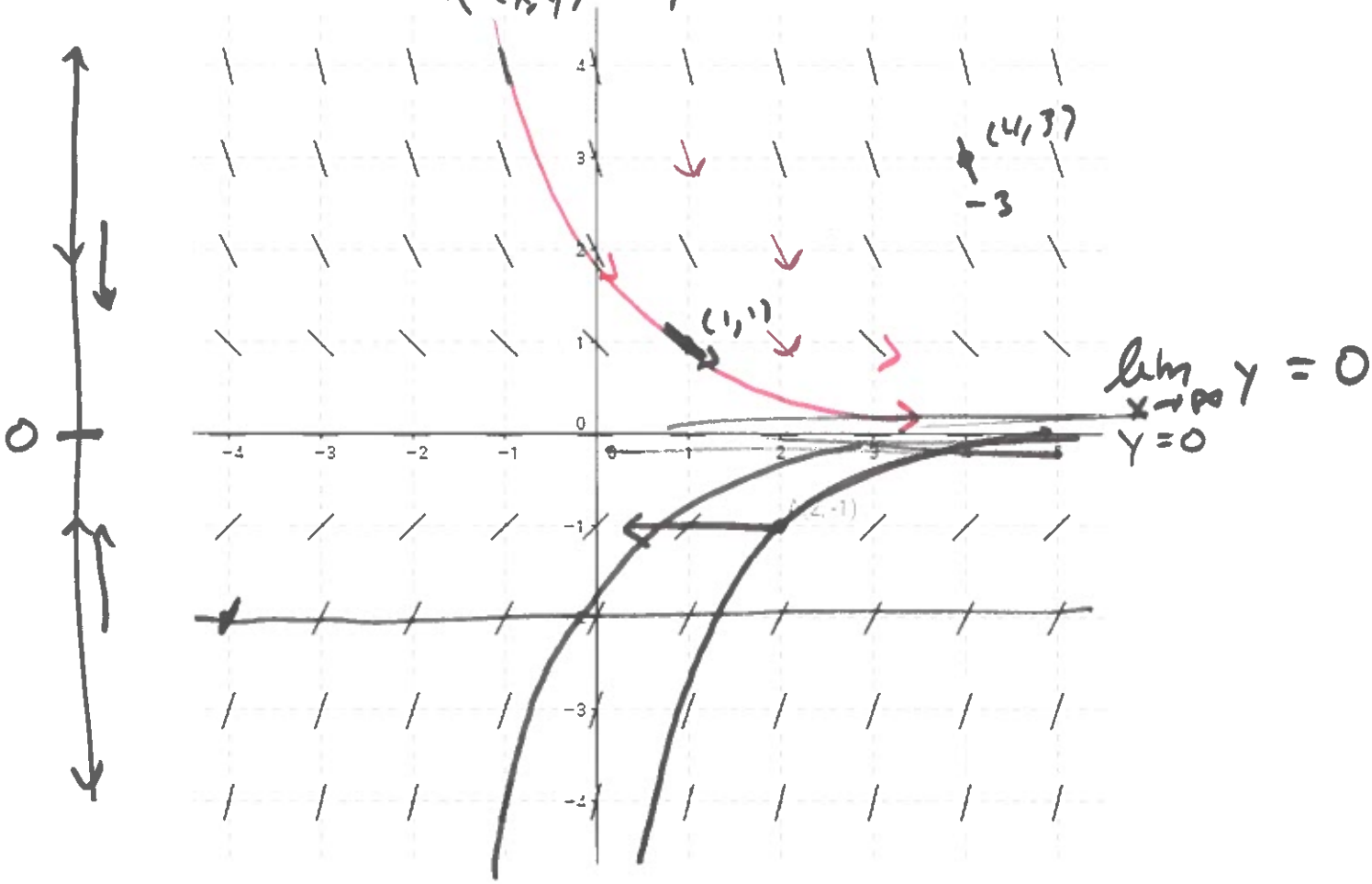
The direction field of a DE describes the flow of solution curves and allows us to determine asymptotic behavior $(\lim_{x\to\pm\infty} y(x))$.

A single solution curve that passes through a point in the direction field must follow the flow pattern of the field.

$\frac{dP}{dt} = kP \qquad ae^{kt} = P$

ex. $\frac{dy}{dx} = -y$ ← no x's

$f(x,y) = -y \qquad f(1,1) = -1$



claim: $y(x) = Ce^{-x}$

Note: A differential equation in the form $\frac{dy}{dx} = f(y)$ in which the independent variable is missing from the right side is called **autonomous.** The slopes corresponding to two different points having the same $y-$ coordinate must be equal. If we know one solution to an autonomous DE, we can obtain infinitely many others just by shifting the graph of the known solution to the right or left.

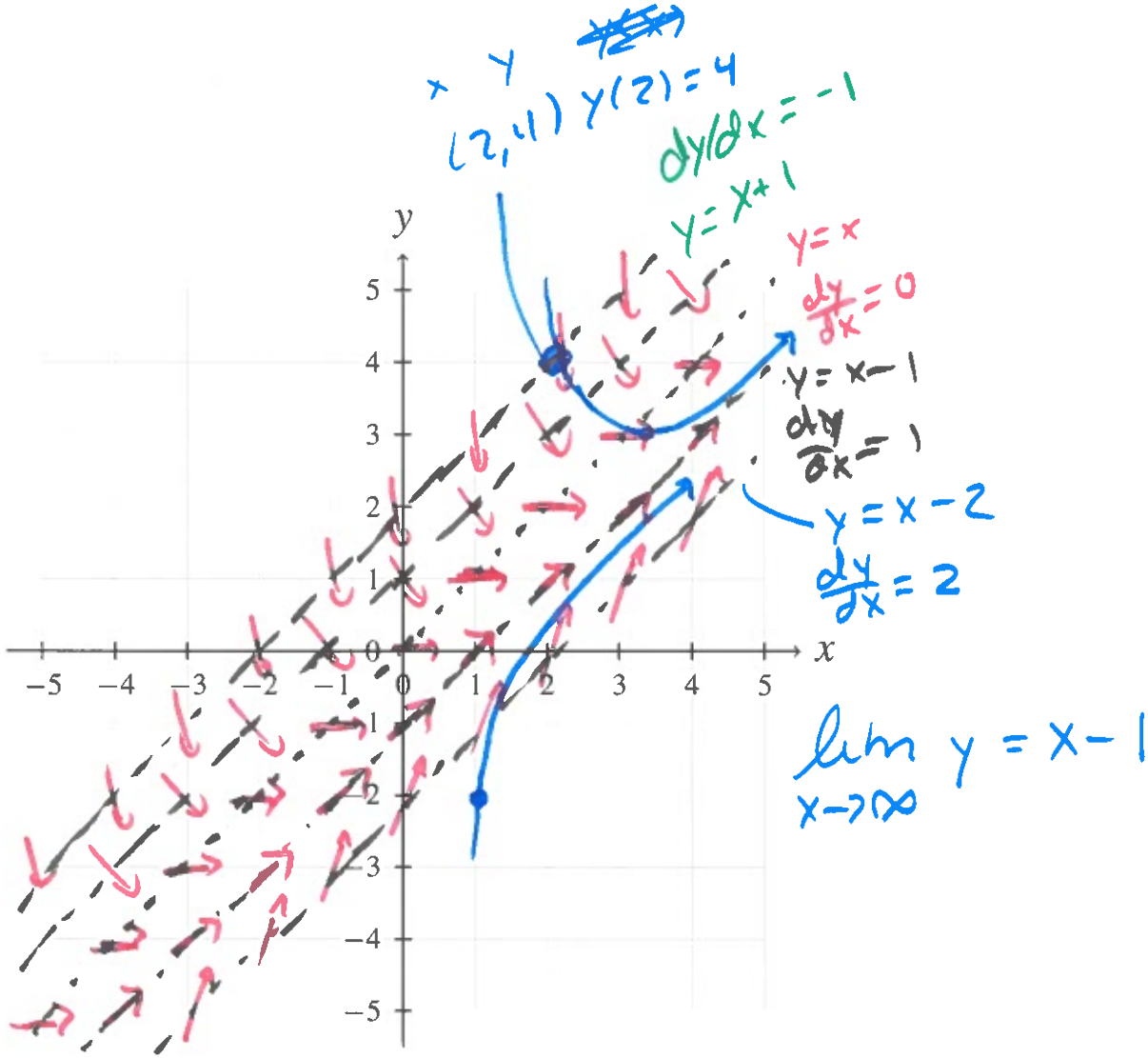
The previous direction field was easy to graph. The slopes were constant on horizonal lines. How would you plot a more difficult example?

Use the **Method of Isoclines**. An isocline is a set of points in a direction field where all solutions have the same slope.

We plot the direction field by determining isoclines.

ex. $\frac{dy}{dx} = x - y$ order 1 linear $\frac{dy}{dx} + y = x$

(1) $\frac{dy}{dx} = 0$ substitute	$x - y = 0$	$y = x$
$\frac{dy}{dx} = 1$	$x - y = 1$	$y = x - 1$
$\frac{dy}{dx} = 2$	$x - y = 2$	$y = x - 2$
$\frac{dy}{dx} = -1$	$x - y = -1$	$y = x + 1$
$\frac{dy}{dx} = -2$	$x - y = -2$	$y = x + 2$



All solutions approach $y = x - 1$ as $x \to \infty$

and as $x \to \infty$ $y \to \infty$

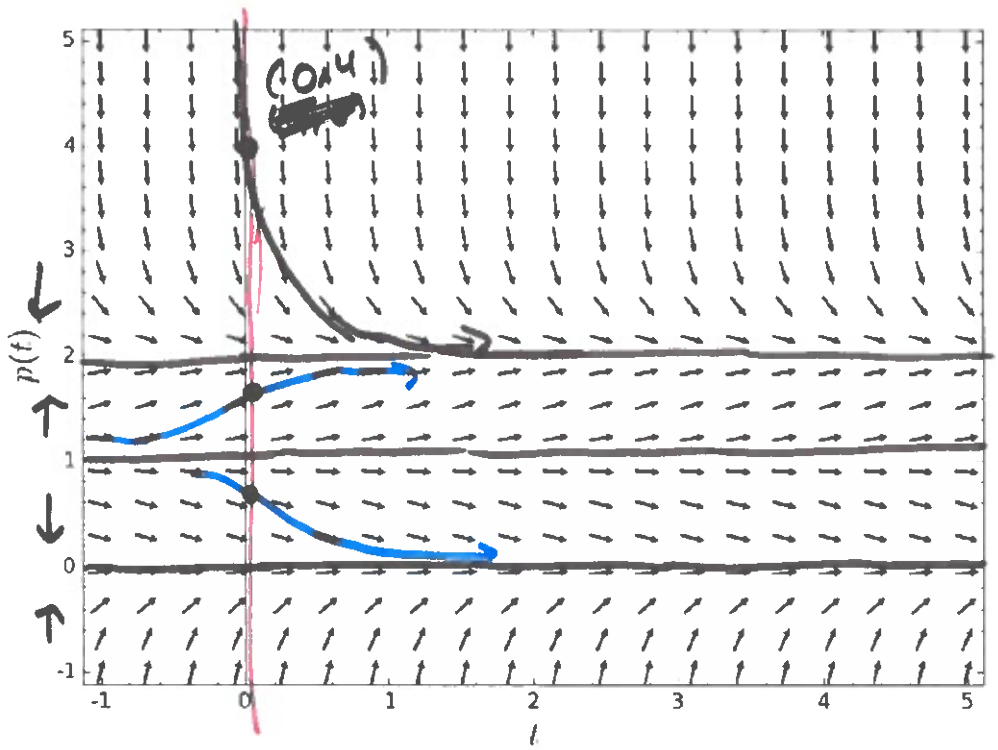
claim: $y = (x-1) + Ce^{-x}$

$$\frac{dP}{dt} = kP$$

ex. The logistic equation for the population p in thousands for a certain species at time t is given by

$$\frac{dp}{dt} = -p(p-1)(p-2).$$ autonomous

The direction field is pictured below.



a. If the initial population is 4000, what can you say about $\lim_{t\to\infty} p(t)$? $=2$

approaches 2000

b. If $p(0) = 1.7$ what is $\lim_{t\to\infty} p(t)$? $=2$

2000

c. If $p(0) = 0.8$ what is $\lim_{t\to\infty} p(t)$? $=0$

0

d. Can a population of 900 ever increase to 1100?

No

Activity

1. Use a computer algebra program like GeoGebra to sketch the direction field for the following DEs:

(a) $\frac{dy}{dx} = \sin x \sin y$

(b) $\frac{dy}{dx} = x^2 + y^2$

2. Sketch the direction field and solution curves that satisfy the given initial conditions:

(a) $\frac{dy}{dx} = x/y, \;\; y(0) = -1$

(b) $\frac{dy}{dx} = 2x^2 - y, \;\; y(0) = 0$

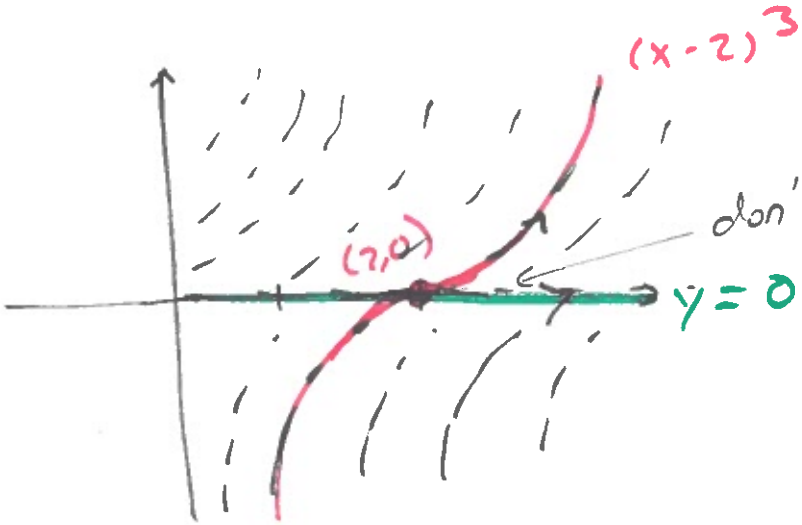
3. Consider $\frac{dy}{dx} = 3y^{2/3}$, $y(2) = 0$

$f(x,y) = 3y^{2/3}$

$f_y = 2y^{-1/3}$ not guaranteed a unique solution

in fact $y=0$ is a solution <u>and</u>

$y = (x-2)^3$ is a solution



don't know which curve to follow at (2,0) since slope is 0.

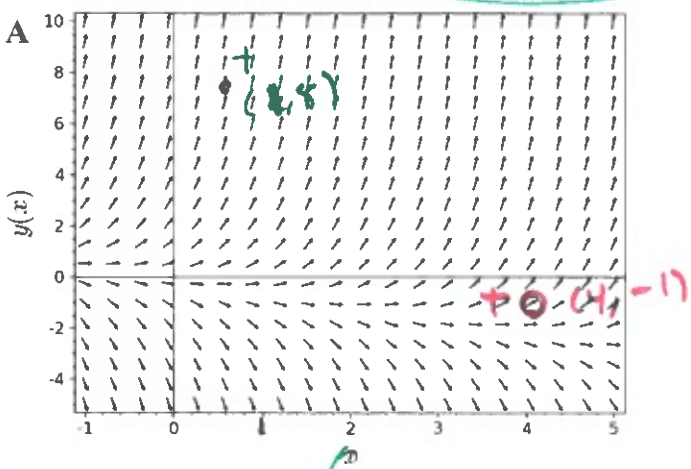
4. Match the IVP to the direction field and sketch the curve passing through the given point.

1. $\dfrac{dy}{dx} = \dfrac{x}{y}$, $y(0) = -1$ C

2. $\dfrac{dy}{dx} = 2x^2 - y$, $y(0) = 0$ B

3. $\dfrac{dy}{dx} = x + 2y$, $y(0) = 1$ A

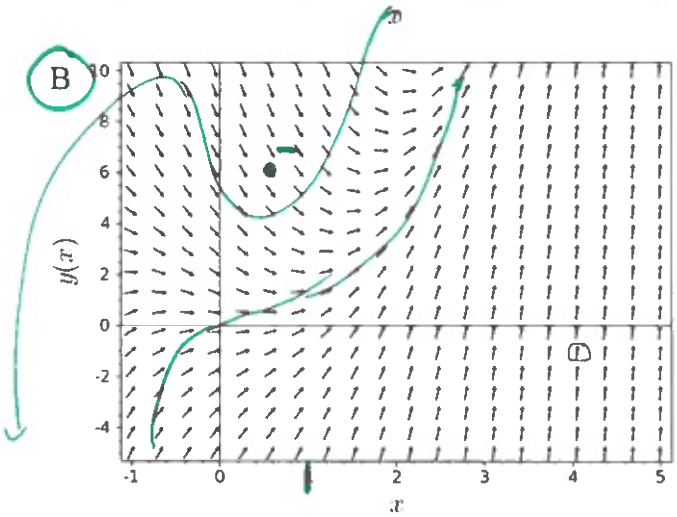
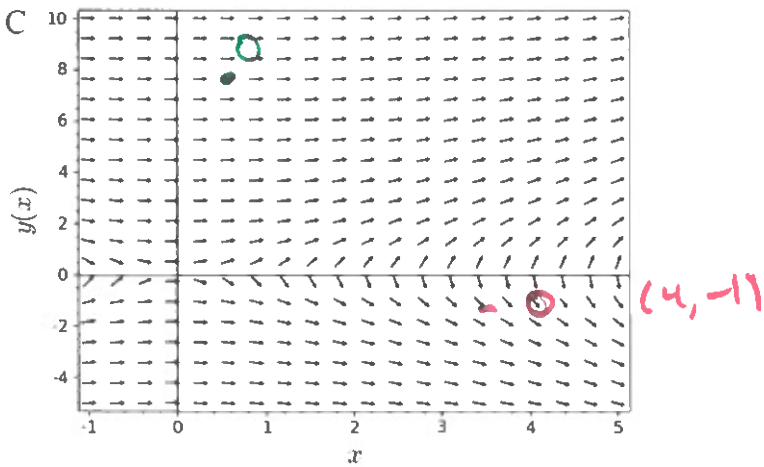
A



1. $\dfrac{dy}{dx} = \dfrac{4}{-1} = -4 < 0$

3. $\dfrac{dy}{dx} = 4 - 2 = 2 > 0$

B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